

TrES-2b

R-1817

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_160806 · created 2026-08-03T04:18:34+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457606.73171 +/- 0.00045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.047 +/- 0.04
Transit depth (Rp/Rs)^2	0.22 +/- 0.38 [%]
Orbital Inclination (inc)	86.95 +/- 2.96
Airmass coefficient 1 (a1)	0.993 +/- 0.016
Airmass coefficient 2 (a2)	0.009 +/- 0.039
Scatter in the residuals of the lightcurve fit is	1.65 %
Best Comparison Star	#3 - [156, 219]
Optimal Aperture	9.28
Optimal Annulus	18.52
Transit Duration (day)	0.084 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.047 ± 0.04 vs 0.1254 (deviation 1.15σ)
Duration fit vs published	0.084 d vs 0.0746 d (deviation 0.19σ)
Mid-time O–C	0.4 min
Depth SNR	0.7
Residual scatter	1.65 %

Light curve

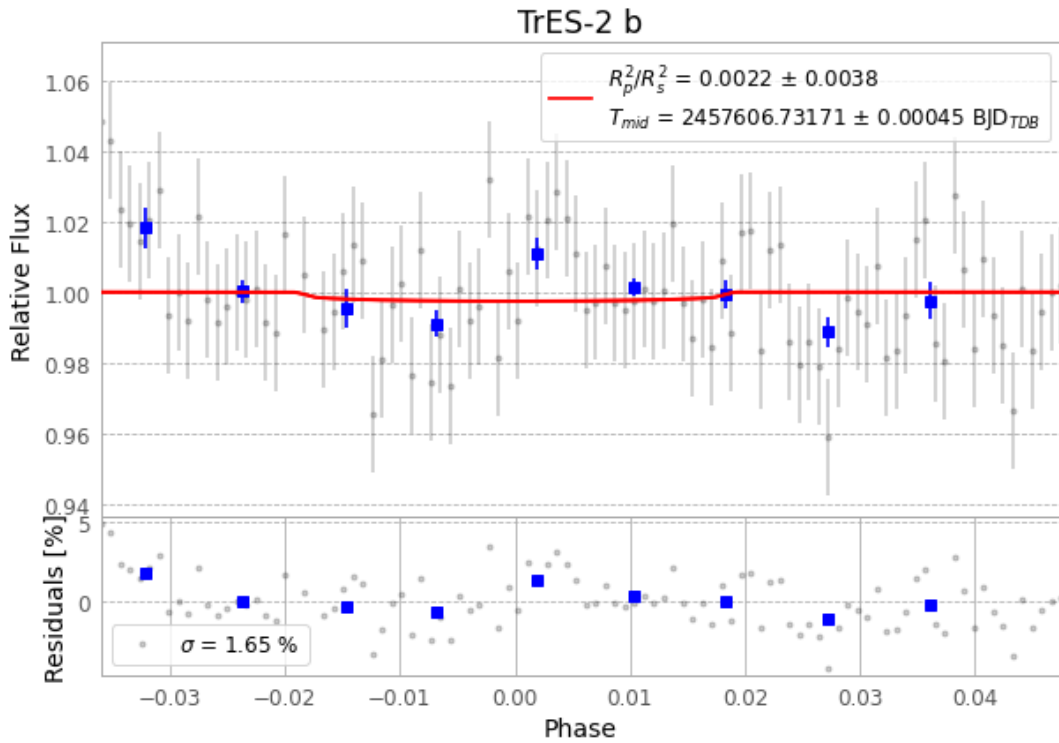


Figure 1 — FinalLightCurve_TrES-2 b_2016-08-05.png (SHA-256 bc5dc23d1a824923c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [93, 377], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 959e1e949b6415ce...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

100 FITS frames (66 MB), TRES-2160806032112.FITS → TRES-2160806081812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-160806.log (5d0ba11428d4...), FinalLightCurve_TrES-2 b_2016-08-05.png (bc5dc23d1a824...), FinalLightCurve_TrES-2 b_2016-08-05.csv (96efa6897dd7...)

Compute resources

Wall time 390.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 04:18Z.